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15 January 1988
SUPERSEDING
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28 June 1984

MILITARY SPECIFICATION

RAINCOATS, MEN'S, QUARPEL

This specification is approved for use by the U.S. Army Natick Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers men's raincoats.

1.2 Classification. The raincoats shall be of one type and class in the following sizes as specified (see 6.2):

Class 1 - Deleted (see 6.7)
Class 2 - Blue 358

Schedule of sizes

Extra short	Short	Regular	Long	Extra long
--	30	30	--	--
32	32	32	32	--
34	34	34	34	34
36	36	36	36	36
38	38	38	38	38
40	40	40	40	40
42	42	42	42	42
--	44	44	44	44
--	46	46	46	46
--	--	48	48	48

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8405

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
V-B-871	-	Button, Sewing Hole, and Button, Staple, (Plastic)
V-T-276	-	Thread, Cotton
V-T-280	-	Thread, Gimp, Cotton, Buttonhole
V-T-295	-	Thread, Nylon
CCC-C-438	-	Cloth, Buckram, Woven and Nonwoven
DDD-L-20	-	Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-B-1860	-	Buckle, Slide, Plastic
MIL-T-3530	-	Thread and Twine: Mildew Resistant or Water Repellent Treated
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43482	-	Cloth, Poplin, Cotton and Polyester (Water Repellent)
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets
MIL-STD-1494 - Provisions for Evaluating Quality of Raincoats

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

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3.2 Material.

3.2.1 Basic material. The basic material shall be a cotton and polyester water-repellent poplin cloth conforming to MIL-C-43482. The cloth shall be dyed Blue 358.

3.2.2 Interlining. The material for the belt interlining shall be cotton buckram cloth, 5.7 ounce, natural or bleached, conforming to type I, class 2 of CCC-C-438.

3.2.3 Fusible interlining. As an alternate to basic material interlinings (the pocket welt, facing, top collar, top collarstand, and shoulder loop interlining), either a black preshrunk 100 percent cotton plain weave cloth or black spunlaced polyester non-woven fabric that have uniformly distributed dot systems of a polyvinyl chloride (PVC)/polyvinyl acetate type, fusible type, fusible adhesive applied to one side in accordance with good commercial practice shall be used. The fusible interlining shall meet the applicable requirements of table I, when tested as specified in 4.2.1.1.

TABLE I. Fusible interlining physical requirements

Type	Weight oz/sq yd	Yarns per inch, min		Breaking strength, lb. minimum	
		Warp	Filling	Warp (machine direction)	Filling (cross machine direction)
Cotton, plain weave	3.3 to 4.1	60	52	36	14
Polyester, spunlaced	2.7 to 3.2	--	--	25	10

3.2.3.1 Colorfastness. All dyed finished fusible interlining materials shall show fastness to laundering (after 3 cycles), equal to or better than the standard sample when tested as specified in 4.2.1.1. When no standard sample is available, the finished fusible materials shall show "fair" fastness to laundering (after 3 cycles).

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3.2.3.2 Fusing conditions. The interlining shall be fused to the basic material on an electric fusing press. For optimum results with a particular fusible style and press used, specific settings should be obtained from the manufacturer of the fusible interlining. For guidance relative to fusing conditions see 6.4.

3.2.3.3 Bond strength. The interlining after fusing to basic cloth, as specified in 3.2.3.2, shall meet the bond strength requirements of table II when tested as specified in 4.2.1.1.

TABLE II. Bond strength requirements

Bond strength, minimum	Ounces per inch width
Initial	20
After 3 launderings	17
After 3 dry cleanings	17

3.2.3.4 Shrinkage. The interlining after fusing to basic cloth, as specified in 3.2.3.2, shall not exceed 1.0 percent shrinkage and shall meet the following requirements when tested as specified in 4.2.1.1.

Shrinkage, percent (max)	
Warp (machine direction)	3.0
Filling (cross machine direction)	2.0

3.2.3.5 Raincoat fusible interlining bond strength. When a fusible interlining is used, the finished raincoat shall meet a minimum bond strength of 17 ounces per inch in the fused areas when tested as specified in 4.2.3. There shall be no "strike through" of adhesive after fusing.

3.2.4 Seam web. The material for fusing the center back, side seams, and sleeve outseams shall be a polyamide fusible web, 3/8 ± 1/16 inch in width and weighing 1.0 ± 0.2 ounces per square yard when tested as specified in 4.2.1.1.

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3.2.5 Thread and gimp.

3.2.5.1 Thread. The thread shall be cotton, type IA3, conforming to V-T-276, ticket nos. 50, 3 ply and 70, 2 ply for seaming and stitching except for sleeve setting, bartacks and top thread of buttonholes which shall be 30, 3 ply. When an automatic button sew and shank type machine is used, the thread shall be waxed nylon monocord, special No. 4, having a minimum breaking strength of 11 pounds and a minimum length per pound of 3250 yards, and shall be tested for these requirements and for colorfastness (see 3.2.5.3) in accordance with test methods outlined in V-T-295. Polyester cotton-covered thread conforming to MIL-T-43548 may be used in lieu of the cotton thread. If polyester cotton-covered thread is used in lieu of cotton thread, a 2-ply may be substituted for 3-ply cotton.

3.2.5.2 Gimp. The gimp for reinforcing the buttonholes shall be cotton No. 8 conforming to type I or II of V-T-280.

3.2.5.3 Colorfastness. The thread and gimp shall be Blue AB, C.A. 66044. The dyed thread shall show fastness to laundering (after 3 cycles) and light. The dyed gimp shall show colorfastness to laundering (after 3 cycles). The thread shall be water repellent treated in accordance with type II, class 2 of MIL-T-3530. Lubrication of the quarpel treated thread by any means in the end item fabricator's plant, prior to or during sewing, is prohibited.

3.2.6 Labels. Each raincoat shall have a class 2 size label and either a class 15 combination identification and instruction label or separate class 1 identification and class 3 instruction label conforming to type VI of DDD-L-20. The labels shall show fastness to laundering. The instruction label or combination label, as applicable, shall include the following information:

RAINCOAT, MEN'S, QUARPEL

Laundering (Preferred Method)

Spot wash badly soiled areas.
Wash in hot water (140°F) using detergent.
Rinse at least 3 times.
Machine spin or squeeze by hand.
Machine dry at moderate heat or hang to dry.
Press with moderately hot iron (synthetic setting).

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Dry Cleaning (Alternate Method)

Dry cleaning in "pure" solvent will not reduce water resistance nor will dry cleaning with detergent providing the detergent is removed by rinsing. If poor water resistance is noted after cleaning, rinsing in water followed by dry cleaning and pressing will renew water resistance.

3.2.7 Buckle. The buckle shall be slide-type, size 2-1/4 inch, conforming to class 3 of MIL-B-1860. The color shall be DOD Blue BU, Cable No. 62002.

3.2.8 Buttons. The buttons shall conform to type II, class D, glossy finish, style 20, 24-line and 45-line, and style 15, 18-line (stay buttons) of V-B-871. The color shall be DOD Blue BU, Cable No. 62002.

3.3 Design. The raincoat is a six-button, double-breasted, belted, trench-type model, with a convertible collar, two-way access hanging pockets, set-in sleeve with tabs, shoulder loops, a right front storm shield, and a storm tab on lower left front.

3.4 Patterns. Standard patterns which show size, directional lines, placement marks, and notches for assembly will be furnished by the Government. The Government patterns shall not be altered in any way and shall be used as a guide for cutting the working patterns. The working patterns shall be identical to the Government patterns. Unless otherwise indicated in table IV, the seam allowances shall be 1/2 inch for top seam of sleeves, side seams, center-back seam, and facing to yoke (lining) seam; 1/4 inch for back pleat insert, shoulder loops, sleeve tabs, welt to coat, and hanging pockets; and 3/8 inch for all other seams.

3.4.1 Pattern parts. The component parts shall be cut from the material specified in table III in accordance with the pattern parts indicated.

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TABLE III. Pattern parts

Material	Pattern nomenclature	Cut parts
Cotton and polyester poplin cloth	Front	2
	Back	2
	Front sleeve	2
	Back sleeve	2
	Top collar	1
	Undercollar	1
	Collarstand	2
	Front facing	2
	Top shield	1
	Under shield	1
	Shoulder loop (and interlining)	6
	Sleeve tab	4
	Storm tab	2
	Pocket welt	2
	Top pocket	2
	Under pocket	2
	Belt	1
	Back yoke	1
	Front yoke	2
	Back pleat facing	1
	Lapel interlining	2
	Undercollar interlining	1
	Under collarstand interlining	1
	Buttonhole reinforcement	1
Cotton, buckram	Belt interlining	1
Cloth, 100 percent cotton plain weave fusible or cloth, spunlaced polyester non-woven fusible or cotton and polyester poplin cloth	Top collar interlining	1
	Top collarstand interlining	1
	Pocket welt interlining	2
	Front facing interlining	2
	Shoulder loop interlining	2

NOTE: Patterns are not furnished for belt keeper, belt loops, hanger, belt safety strap, facing for pocket and hand-openings, and binding for armholes. See table IV for cutting directions.

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3.5 Construction. The construction shall conform in all respects to the requirements specified in table IV and herein. Figure 1 is furnished solely for guidance and information. Should variation from the document appear in figure 1, the document shall govern.

3.5.1 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table IV. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the chain or underside of the stitch shall be on the inside of the raincoat.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/2 inch minimum except when caught in other stitching or turned under in a hem. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin, top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

3.5.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the end item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area and continue a minimum of 1/2 inch beyond the defective area on to the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Type 401 stitching. Thread tension shall be maintained so that there will be no loose stitching or excessively tight stitching resulting in puckering of the materials sewn. Both ends of all seams or stitching produced with 401 stitch type, when not turned under in a hem or held down by other stitching, shall have a 1/2 to 3/4 inch chain extending beyond each end. The looper (under thread) shall be hidden between the cover and lining assembly. All repairs shall be in accordance with 3.5.1.1.1a and 3.5.1.1.1b. Repairs may be made using 301 stitch type.

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3.5.2 Bartacking. Bartacking shall be 3/8 to 1/2 inch long and 1/16 to 3/32 inch wide. Bartacks shall be free from thread breaks and loose stitching.

3.5.3 Buttonholes. The buttonholes shall be eyelet-end, square-bar, cut-first type, worked over gimp with the ends of the gimp pulled through the underside. The end shall be bartacked in a separate operation.

3.6 Manufacturing operations requirements. The raincoats shall be made in accordance with the operation requirements specified in table IV. The manufacturer is not required to follow the exact sequence of operations unless otherwise specified. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished raincoat.

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TABLE IV. Manufacturing operations requirements

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
1.	Cutting					
	a. Cut all parts of the raincoats in strict accordance with patterns furnished, which show directional lines, size, shape, placement of pockets, loops and tabs, and notches for proper assembly of all parts. Directional lines shall be placed in the warp direction. The pocket opening may be die-cut, but not tongue-notched, during the cutting operation.					
	b. Cut all basic material parts of the raincoat from the same piece of material except under and top pocket; front and back yoke (linings); under shield; under part of shoulder loop; under part of sleeve tab; storm tab; all basic material interlinings; buttonhole reinforcement piece; belt keeper; belt loops; hangers; belt safety strap; facing for pocket hand openings; armhole binding; and storm tab and button reinforcement pieces which may be cut from ends or as an alternate cut all fusible interlinings for topcollar interlining, top collarstand interlining, pocket welt interlining, front facing interlining and shoulder loop interlining. The shade of the belt keeper, belt safety strap, belt loops, back and front yoke (linings) and under and top pockets shall approximate the shade of the raincoat. When cut from ends, back and front yoke (linings) and under pockets shall match each other.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/looper
1.	Cutting - continued					
	c. Cut the following basic material parts in the warp or filling direction:					
	(1) Stripping for belt keeper, belt loops, and hanger sufficiently wide to finish 3/8 ± 1/16 inch (see operation 4).					
	(2) Stripping for belt safety strap sufficiently wide to finish 5/8 to 3/4 inch (see operation 5).					
	(3) Facings for pocket hand-openings (one each opening) 1-1/2 to 1-5/8 inches wide by 9-1/2 to 9-3/4 inches long (see operation 16).					
	(4) Storm tab and button reinforcement pieces of sufficient size to reinforce storm tab at joining seam and sleeve button (see operations 21 and 26).					
	d. Cut the binding for the armholes from basic material, on the bias, 1 to 1-1/4 inches wide (see operation 23).					
2.	Replacement of damaged parts					
	During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1494 shall be removed from production and replaced with non-defective and properly matched components.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
3.	Shade and size marking					
	Mark, ticket, or bundle all component parts of the raincoat (except those parts cut from ends) to ensure a uniform shade and size throughout the garment.					
4.	Make belt keeper, belt loops and hanger (see operation 1.c.(1))					
	a. Fold stripping in half lengthwise, turn in both raw edges and stitch with two rows of stitching 1/16 to 1/8 inch from each edge.	301 or 401	EFp-2	9-11	50-3 50-3	50-3 50-3
	b. Cut stripping in lengths as follows:					
	(1) Belt keeper - 5-3/4 to 6 inches.					
	(2) Belt loops (three each raincoat) 3-3/4 to 4 inches.					
	(3) Double belt loop (one each raincoat) 6-7/8 to 7 inches.					
	(4) Hanger loop - 2-3/4 inches.					
5.	Make belt safety strap (see operation 1.c.(2))					
	a. Fold stripping in half lengthwise; turn in both raw edges and stitch with one or two rows of stitching 1/16 to 1/8 inch from (each) edge.	301 301	EFp-1 or EFp-2	9-11 9-11	50-3 50-3	50-3 50-3
	b. Cut strap 7-1/2 to 7-3/4 inches long.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
5.	Make belt safety strap (see operation 1.c.(2)) - continued					
	c. Turn one end of strap back with raw edge turned under, stitch 1/8 to 3/16 inch from edge	301	EFb-1	9-11	50-3	50-3
	or					
	d. Raw edges at one end may be turned to inside and stitch 1/16 to 1/8 inch from the turned edges.	301	SSc-1	9-11	50-3	50-3
	e. Make a 5/8 ± 1/16 inch finished cut length buttonhole at the finished end of the strap with the inside edge of the eyelet 5/8 to 3/4 inch from end. Buttonhole shall not be off center by more than 1/16 inch.	Buttonhole		50-52 per buttonhole	30-3	50-3
6.	Make sleeve tabs.					
	Finished appearance. The sleeve tabs shall finish flat and smooth and be uniform in appearance.					
	a. Superimpose plies of sleeve tabs and seam together, leaving end to be attached to sleeve open.	301 or 401	SSae-2(a)	9-11	50-3 50-3	50-3 70-2
	b. Trim corners, turn, work out edges and corners, and stitch 3/8 to 1/2 inch from turned edges.	301	SSae-2(b)	9-11	50-3	50-3
	c. Make a 5/8 to 3/4 inch finished cut length buttonhole, centered on tab at mark indicated on pattern (3/16 inch off-center tolerance) with the inside edge of eyelet 3/4 ± 1/8 inch from finished edge.	Buttonhole		50-52 per buttonhole	30-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
7.	Make storm tabs.					
	Finished appearance. The storm tabs shall finish flat and smooth.					
	a. Superimpose plies of storm tab and seam together leaving end to be attached to facing open.	301 or 401	SSae-2(a)	9-11	50-3 50-3	50-3 70-2
	b. Trim point, turn, work out edges and point, and stitch 3/8 to 1/2 inch from turned edges.	301	SSae-2(b)	9-11	50-3	50-3
	c. Make a 5/8 to 3/4 inch finished cut length buttonhole, centered on tab at mark indicated on pattern (1/8 inch off-center tolerance) with the inside edge of eyelet 5/8 ± 1/8 inch from finished edge.	Buttonhole		50-52 per buttonhole	30-3	50-3
8.	Make shoulder loops (right and left)					
	Finished appearance. The shoulder loops shall finish flat and smooth and be uniform in appearance.					
	a. Superimpose plies of shoulder loops on shoulder loop interlining and join all plies leaving end to be attached to shoulder, open. When alternate interlining (fusible material) is used, position fusible interlining to top shoulder loops and fuse.	301 or 401	SSae-2(a)	9-11	50-3 50-3	50-3 70-2
	b. Trim points, turn, work out points and edges, and stitch 3/8 to 1/2 inch from turned edges.	301	SSae-2(b)	9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
8.	Make shoulder loops (right and left) - continued					
	c. Make a 5/8 to 3/4 inch finished cut length buttonhole, centered on shoulder loop at mark indicated on pattern (1/8 inch off-center tolerance) with the inside edge of eyelet 7/8 ± 1/8 inch from point.	Buttonhole		50-52 per buttonhole	30-3	50-3
9.	Make front shield					
	a. Stitch front shields together along front and bottom edges.	301 or 401	SSae-2(a)	9-11	50-3 50-3	50-3 70-2
	b. Trim corners, turn, work out edges, and stitch 3/8 to 1/2 inch from edge.	301	SSae-2(b)	9-11	50-3	50-3
	c. Make a 5/8 to 3/4 inch finished cut length diagonal buttonhole at mark as indicated by pattern with eyelet end toward corner edge and purling on outside of shield. Inside edge of eyelet shall finish 3/4 to 7/8 inch from edge.	Buttonhole		50-52 per buttonhole	30-3	50-3
10.	Make belt					
	Finished appearance. The belt shall finish flat and smooth and the gage of stitching shall be uniform in width. The belt shall finish 2-1/4 + 1/16 - 1/8 inches wide. The purling of the buttonhole on the belt safety strap shall finish away from belt.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
10.	Make belt - continued					
	a. Position the belt keeper across the width of the belt and the raw end of the belt safety strap on pattern. Belt keeper and safety strap may be tacked into position. Fold belt in half lengthwise with the belt keeper and safety strap enclosed within the fold. Position the raw edges of the folded belt on the interlining and stitch belt to interlining so that the line of stitching will be at the center of the interlining.	301 or 401		9-11	50-3 50-3	50-3 70-2
	b. Shift position of belt on interlining so that belt and interlining are superimposed and points coincide, and stitch around pointed end.	301 or 401	SSae-3(a)	9-11	50-3 50-3	50-3 70-2
	c. Trim point and corners; turn belt to outside; force out point and edge stitch both sides and point of belt with two rows of stitching, 3/8 to 1/2 inch apart with outside edge stitch 1/16 to 3/32 inch from edge. The belt looper shall not be caught in the edge stitching.	301	OSf-2 and SSae-3(b)	9-11	50-3	50-3
	d. Thread straight end of belt through buckle; fold back 2-1/4 ± 1/4 inches; turn under the raw edge and stitch with two rows of stitching, adjacent to each other, 1/16 to 1/8 inch from turned edge.	301	EFb-2	9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
11.	Make collar					
	a. Stitch undercollar interlining to undercollar 1/8 ± 1/16 inch from edge around all edges and quilt the interlining material to the undercollar with two rows of stitching. Stitch the first row approximately 1-1/2 inches from the outer edge of the undercollar and the second row 3/4 inch from the first row.	301 or 401	SSa-1	9-11	50-3 50-3	50-3 70-2
	b. Stitch under collarstand interlining to under collarstand 1/8 ± 1/16 inch from edge around all edges.	301 or 401	SSa-1	9-11	50-3 50-3	50-3 70-2
	c. Stitch undercollar to under collarstand, turn seam toward stand and raise stitch 1/16 to 3/32 inch from edge.	301 or 401 and 301	LSq-2(a) LSq-2(b)	9-11 9-11	50-3 50-3 50-3	50-3 70-2 50-3
	d. Stitch top collar to top collarstand, turn seam toward stand and raise stitch 1/16 to 3/32 inch from edge.	301 or 401 and 301	LSq-2(a) LSq-2(b)	9-11 9-11	50-3 50-3 50-3	50-3 70-2 50-3
	e. Join top collar to undercollar along top and side edges, trim points, turn, work out points and edges, and edge stitch 3/8 to 1/2 inch from edge.	301 or 401 and 301	SSae-2(a) SSae-2(b)	9-11 9-11	50-3 50-3 50-3	50-3 70-2 50-3
	f. Stitch top collar to undercollar through all plies along bottom edge 1/8 to 3/16 inch from edge.	301		9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
11.	Make collar - continued					
	When alternate interlining is used					
	g. Position fusible interlining to top collar and fuse. Stitch undercollar to top collar around the sides and edges.	301 or 401	SSe-2(a) SSe-2(a)	10-12 10-12	50-3 50-3	70-2 70-2
	h. Trim, turn, and force out edges and points and stitch 3/8 to 1/2 inch from edge.	301	SSe-2(b)	9-11	50-3	50-2
	i. Position fusible interlining to under collarstand and fuse.					
	j. Position undercollar between top and under collarstand, and with raw edges even stitch along side and top edge of collarstand.	301	SSe-2(a)	10-12	50-3	70-2
	k. Turn collarstand to finished position forcing out points and stitch 1/16 to 3/32 inch from turned edges.	301	SSe-2(b)	10-12	50-3	50-3
12.	Make pocket welts					
	a. When alternate interlining (fusible material) is used, position fusible interlining to underside of top portion of welt and fuse.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
12.	Make pocket welts - continued					
	b. Fold welt in half lengthwise and stitch across width at both ends.	301 or 401	SSae-3(a)	9-11	50-3 50-3	50-3 70-2
	c. Turn, work out points and edges, and stitch the ends and folded side 3/8 to 1/2 inch from edge.	301	SSae-2(b) and OSf-1	9-11	50-3	50-3
13.	Hem yokes and attach to facings					
	a. Position size label on back yoke, to the right side of pleat (as worn) 1 ± 1/4 inches from pleat and 1/2 ± 1/4 inch below top joining seam and stitch on all four sides.	301	LSbj-1	9-11	50-3	50-3
	b. Hem bottom of front and back yokes with raw edge turned in 3/16 to 1/4 inch and stitch 1/16 to 3/32 inch from edge. Hem shall finish 1/4 to 3/8 inch wide.	301	EFb-1	9-11	50-3	50-3
	c. Superimpose front facing on lapel interlining with front yoke positioned between facing and interlining as indicated by marks on pattern and stitch yoke to facing and interlining.	301 or 401	SSq-2(a)	9-11	50-3 50-3	50-3 70-2
	d. Turn facing and interlining to front and raise stitch, 1/16 to 3/32 inch from turned edge.	301	SSq-2(b)	9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
13.	Hem yokes and attach to facings - continued					
	e. Slash seam allowance of front facings at bottom of yoke, 1/4 inch as indicated by mark on pattern, fold under back raw edges of facings and stitch back edge forming 1/4 ± 1/16 inch hem.	301	EFb-1	9-11	50-3	50-3
	f. Position buttonhole reinforcement piece to left side of coat only, overlapping bottom edge of left lapel interlining 1/2 to 5/8 inch, with front edge of reinforcement with the front edge of facing in line with front edge of lapel interlining. Stitch lapel interlining, 1/16 to 3/32 inch from edge of interlining, to facing along top edge and down front edge, continuing the left front stitching to bottom edge of buttonhole reinforcement. (Top edge of buttonhole reinforcement may be pretacked to lapel interlining prior to stitching to facing.)	301 or 401	SSa-1	9-11	50-3 50-3	50-3 70-2
	g. Join back yoke to front yokes at shoulders. The seam allowances shall be pressed open or opened when joining sleeves and collar.	301 or 401	SSa-1	9-11	50-3 50-3	50-3 70-2
	NOTE: When alternate interlining fusible material is used, delete lapel interlining (basic material).					
14.	Make sleeves					
	Finished appearance. The sleeves shall be uniform in appearance and width. The seams shall be pressed smooth and flat. The sleeve tabs shall be properly positioned and parallel to the bottom edge of sleeve.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
14.	Make sleeves - continued					
	a. Join front sleeve to back sleeve at top arm seam with edge of front sleeve positioned 1/2 inch behind edge of back sleeve, turn back sleeve seam allowances over front sleeve and stitch 1/16 to 3/32 inch from edge of back sleeve.	301	SSw-2(a)	9-11	50-3	50-3
	b. Turn seam toward front of sleeves and stitch 1/16 to 3/32 inch from inside of folded edge.	301	SSw-2(b)	9-11	50-3	50-3
	c. Join underarm seam of sleeve with a double-lapped, double-stitched seam, 3/16 to 1/4 inch gage, catching the sleeve tab in the seam as indicated by marks on pattern.	301 or 401	LSc-2	9-11	50-3 50-3	50-3 70-2
	d. Tab may be pretacked to back sleeve prior to joining underarm seam.					
	e. Hem bottom of sleeves with raw edges turned under 3/8 to 7/16 inch and stitch 1/16 to 1/8 inch from edge. Hem shall finish 1 ± 1/8 inches wide.	301	EFb-1	9-11	50-3	50-3
15.	Make pockets and attach labels					
	a. Stitch identification label on all four sides to back side of right under pocket (to face wearer) centered on pocket (1/2 inch tolerance) and 1 ± 1/4 inches below top edge.	301	LSbj-1	9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread Needle	Thread Bobbin/ looper
15.	Make pockets and attach labels - continued					
	b. Position instruction label centered directly below identification label and stitch to pocket on all four sides. Top edge of instruction label may be caught in bottom edge of stitching of identification label.	301	LSbj-1	9-11	50-3	50-3
	c. Combination label (when used) shall be positioned as specified for the identification label (see operation 15.a).					
	d. Slash top pocket opening, superimpose pocket pieces, and stitch together around all edges.	301	SSe-2(a)	9-11	50-3	50-3
		or 401			50-3	70-2
	e. Turn pocket inside out and edge stitch 1/16 to 1/8 inch from edge.	301	SSe-2(b)	9-11	50-3	50-3
	f. Turn under back raw edge of top pocket and stitch to under pocket 1/16 to 3/32 inch from turned edge forming facing.	301	LSd-1	9-11	50-3	50-3
16.	Set pockets					
	Finished appearance. The pockets and welts shall be uniform in appearance and placement and shall lie flat and smooth, without twisting, pleating, or puckering. The stitching shall be uniform in width.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
16.	Set pockets - continued					
	a. Set pocket welt on front as indicated by marks on pattern. Seam welts to fronts 1/4 inch from raw edge. The stitching shall not extend beyond the pocket line.	301 or 401	SSa-1		50-3 50-3	50-3 70-2
	b. Fold facing for hand-opening (see operation 1.c(3)) in half lengthwise, position raw edges of facing on opposite side of pocket opening, and mark with raw edges abutting raw edges of welt. Stitch facing to front along pocket opening mark from top to bottom of opening 1/4 inch from raw edges.	301 or 401			50-3 50-3	50-3 70-2
	c. Slash pocket opening and tongue notch ends. NOTE: Cuts shall not be visible on finished coat.					
	d. Turn facing to inside of raincoat, turn under raw edge and stitch to coat from top to bottom of opening 1/16 to 3/32 inch from folded edge.	301		9-11	50-3	50-3
	e. Seam unfinished side of under pocket to welt-to-front joining seam (This may be performed as part of operation 16.a.).	301	SSa-2	9-11	50-3	50-3
	f. Turn pocket to inside and raise stitch through welt and top pocket 1/16 to 3/32 inch from turned edge.	301		9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
16.	Set pockets - continued					
	g. With pocket in a flat position, stitch down top and bottom edges of welt, through front and hanging pocket, with two rows of stitching. The inner row shall superimpose the 1/2 inch stitching on welt; the outer row shall be 1/16 to 3/32 inch from edge, and the stitching at ends of hand opening shall be diagonal to the corners of the hand opening.	301		9-11	50-3	50-3
17.	Join backs					
	a. Join backs at center back with edge of left positioned 1/2 inch behind edge of right back, turn right back seam allowance over left back, and stitch 1/16 to 3/32 inch from edge of right back and from top of coat to notch at top of pleat. The left back shall be notched at top of pleat so that the left and right folded edges will be butted.	301	SSw-2(a)	9-11	50-3	50-3
	b. Turn seam toward left back and stitch 1/16 to 3/32 inch from inside folded edge.	301	SSw-2(b)	9-11	50-3	50-3
18.	Join pleat facing to back					
	a. Join left and right side of facing to right and left backs at pleat with bottom edges even.	301 or 401	SSae-2(a)	9-11	50-3 50-3	50-3 70-2
	b. Turn seam and stitch, 3/8 to 1/2 inch from folded edges from bottom of coat to top edge of facing.	301	SSae-2(b)	9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
18.	Join pleat facing to back - continued					
	c. Form pleat. At top edge of pleat facing, turn under seam allowance forming a point, and seam to coat along top edges, 1/16 to 3/32 inch from turned edges, pinch and continue directly across width of pleat with two rows of stitching 3/16 to 1/4 inch gage, closing pleat. Point of pleat facing shall be in line with the center back joining seams (± 1/8 inch off-center tolerance).	301		9-11	50-3	50-3
19.	Join front shield to right front					
	Position front shield to right front with top and armhole edges even and stitch 1/8 ± 1/32 inch from edge.	301 or 401	SSa-1	9-11	50-3 50-3	50-3 70-2
20.	Join front to back					
	a. Join fronts to back at side seams with edge of back positioned 1/2 inch behind edge of front, turn front seam allowance over back and stitch 1/16 to 3/32 inch from edge of front.	301	SSw-2(a)	9-11	50-3	50-3
	b. Turn seam towards back of coat and stitch 1/16 to 3/32 inch from inside folded edge.	301	SSw-2(b)	9-11	50-3	50-3
	c. Join fronts to back at shoulder seams with raw edges even and stitch 3/8 inch from edge.	301	LSq-2(a)	9-11	50-3	50-3
	d. Turn shoulder seam allowance to back of coat and stitch 1/8 inch from inside raw edge.	301	LSq-2(b)	9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
21.	Attach storm tab to left facing					
	Position storm tab to facing at marks as indicated on pattern, and stitch to facing and reinforcement, 3/16 to 1/4 inch from edge, turn tab, and raise stitch 1/4 to 3/8 inch from turned edge continuing the stitching across the top edge, terminating 1/2 ± 1/8 inch from corner. A bartack, positioned 1/2 ± 1/8 inch from corner, may be used in lieu of stitching across top edge. The buttonhole purling shall face the wearer on the finished raincoat.	301 or 301 and Bartack	LSbl-2 LSbl-2	9-11 28 per bartack	50-3 30-3	50-3 30-3
22.	Join facings and collar to coat					
	Finished appearance. The lapels and fronts shall finish flat and smooth without twists, pleats, or puckers. The edge stitching shall be uniform in width.					
	a. Stitch facings to fronts along bottom and front edges and along top of lapel to notch at collar.	301 or 401	SSae-2(a)	9-11	50-3 50-3	50-3 70-2
	b. Insert collar between outer shell and facing and back yoke at neck line, with all raw edges even stitch collar to outer shell, facing and back yoke from notch to notch, forming a pleat in center of yoke as indicated by notches on pattern. As an alternative, the collar may be stitched to outer shell prior to joining facings and yoke along neckline.	301		9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
22.	Join facings and collar to coat - continued					
	c. Trim seam allowance at lapel point, turn facing and back yoke to inside, working out all edges and raise stitch facing and back yoke to collar, 1/16 to 3/32 inch from edge, through neck seam allowance only, from one end of collarstand to opposite end of collarstand.	301		9-11	50-3	50-3
	d. Hem bottom of coat with raw edge turned under 1/4 to 3/8 inch and stitch 1/16 to 1/8 inch from edge. Stitching shall end at back edge of facings. Hem shall finish 1 ± 1/8 inches wide. The back pleat facing seam allowances shall be toward the side seams in the finished hem.	301	EFb-1	9-11	50-3	50-3
	e. Edge stitch front to facing, 1/2 ± 1/16 inch from edge, along bottom to back edge of facings and front edges and along top edge of lapel even with notch and continue the stitching perpendicular to top edge of lapel and terminating at collar notch at lapel.	301	SSae-2(b)	9-11	50-3	50-3
	f. Stitch front and back yoke to armhole, 1/8 to 3/16 inch from raw edges.	301	SSa-1	9-11	50-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
23.	Join sleeves to coat					
	a. Set sleeves with top arm seam in line with shoulder seam and underarm seam within 1/4 inch of side seam and stitch to coat distributing fullness. Shoulder loop shall be caught in the stitching at marks as indicated on patterns. Buttonhole purling shall be on the top when loop is buttoned. (The shoulder loop may be pre-tacked to shoulder.)	301	SSa-1	9-11	30-3	30-3
	b. Bind the raw edges of the armhole joining seam with bias cut stripping (see operation 1.d) with the edge of the stripping turned under, and stitch 1/16 to 3/32 inch from turned edge of stripping. Raw ends of stripping shall be folded over and caught in the stitching or folded over and bartacked. Stripping shall finish 1/4 to 3/8 inch wide.	301 or 301 and Bartack	BSc-1 BSc-1	9-11 9-11 28 per bartack	30-3 30-3 30-3	30-3 30-3 30-3
24.	Attach hanger loop and belt loops					
	a. Position hanger loop to center of top collarstand (1/4 inch off center tolerance) with bottom edge of hanger loop adjacent to neck seam with ends of hanger turned under, and stitch or bartack hanger to top collarstand through under collarstand, 1/8 to 3/16 inch from ends. Hanger shall finish 2 to 2-1/4 inches long.	301 or Bartack	LSd-1	9-11 28 per bartack	50-3 30-3	50-3 30-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
24.	Attach hanger loop and belt loops - continued					
	b. Position one belt loop on right side seam as indicated by marks on pattern and one belt loop each front above pocket welts as indicated by marks on pattern with ends of loop turned under and stitch or bartack both ends of loops to coat, 1/8 to 3/16 inch from ends. Loops shall be straight and shall finish 3 to 3-1/4 inches long. Bartack or stitching shall cover width of loop.	301 or	LSd-1	9-11	50-3	50-3
		Bartack		28 per bartack	30-3	30-3
	c. Form a loop, 3 to 3-1/4 inches long with one end folded over and enclosing the other end. Stitch or bartack loop to side seam 1/8 to 3/16 inch from each end. Bartacks shall be 3/8 to 1/2 inch and stitching or bartack shall cover width of loop.	301 or	LSd-1	9-11	50-3	50-3
		Bartack		28 per bartack	30-3	30-3
25.	Make buttonholes					
	Finished appearance. The purling of buttonhole on left and right lapel and the one buttonhole on right front shall finish on the facing side. The purling on all other buttonholes shall finish on outside of coat.					
	a. Make three 1-1/8 to 1-1/4 inches finished cut-length horizontal buttonholes in left front and one, 1-1/8 to 1-1/4 inches finished cut-length horizontal buttonhole in right front at marks as indicated on patterns, with inside edge of eyelets 3/4 to 7/8 inch from finished front edges of coat.	Buttonhole		70-74 per buttonhole	30-3	50-3

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
25.	Make buttonholes - continued					
	b. Make two, 1-1/8 to 1-1/4 inches finished cut-length horizontal buttonholes (one in each lapel), 1-1/2 ± 1/8 inches down and parallel with top finished edge of lapel, with inside edge of eyelets 3/4 to 7/8 inch from finished front edge of coat.	Buttonhole		70-74 per buttonhole	30-3	50-3
26.	Mark and sew on buttons					
	Mark position of buttons and sew a button at each mark as indicated. The 45-line button on the facing side of the left front; the four 45-line buttons on the right front; and the 24-line front shield buttons on the left and right fronts shall be tightly shank-wrapped 1/4 inch unless automatically shanked which shall have a minimum of ten wraps with three locking stitches. The end of the thread on hand sewn buttons shall be tacked off.	301 or 101 or Hand		14-16 per button 14-16 per button 4-6 double thread	50-3 50-3 50-3	50-3
	(1) One 24-line button to underside of belt (toward center) to correspond to buttonhole in belt safety strap. Stitching shall not be exposed on outside of belt.					
	(2) One 24-line button on each shoulder seam through seam and lining to correspond to buttonhole in shoulder loops. Stitching may be through the lining.					
	(3) One 24-line button on right front through front and facing to correspond to buttonhole on front shield.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
26.	Mark and sew on buttons - continued					
	(4) One 24-line button on each front sleeve through sleeve and reinforcement piece to correspond to buttonhole on sleeve tabs.					
	(5) Three 45-line buttons on right front. Top two buttons through all plies and a stay button and bottom button through the front and top ply of the pocket. The buttons shall correspond to buttonholes on left side positioned as indicated by marks on pattern.					
	(6) One 45-line button on right front, through all plies and stay button to correspond to button-hole on left lapel when coat is buttoned at neck.					
	(7) One 24-line button on right front through all plies and stay buttons to correspond with button-hole on storm tab when coat is buttoned.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
26.	Mark and sew on buttons - continued					
	(8) Three 45-line buttons on left front. The top button shall be sewn through all plies and through a 45-line button on facing side with buttons back to back. The center button shall be sewn through all plies and a stay button. The bottom button shall be sewn through the front and top ply of the pocket. The buttons shall be in line with buttonholes on left front (1/4 inch off alignment allowed), positioned as indicated by marks on pattern.					
	(9) One 24-line button on left front, through front and interlining only, shall correspond to buttonhole in front shield when lapel is buttoned.					
27.	Bartack hanging pockets					
	Bartack front of hanging pockets on the inside of coat through back edge of front facing at top and bottom of pocket with vertical bartacks. The facing shall not be distorted when bartacked.	301 or Bartack		9-11 28 per bartack	50-3 30-3	50-3 30-3
28.	Attach belt to coat					
	a. Thread belt between double loops and through belt loops. Buckle shall be on left side of coat (as worn).					
	b. Thread belt safety strap through double belt loop between back loop and coat and button strap to belt.					

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TABLE IV. Manufacturing operations requirements - Continued

No.	Requirement	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
29.	Clean coat Trim ends of threads and remove loose threads. Remove the spots and stains and shade tickets.					
30.	Press coat a. Press fronts, backs, sleeves, collar, belt, and armholes smooth with a heated pressing iron or steam pressing machine. Crease press front and back sleeve, half way up with seams superimposed. b. Press back pleats so that the folds of the pleats abut at the top and bottom.					

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3.7 Finished measurements. The finished dimensions for the coat shall conform to measurements for each respective size, as listed in table V.

TABLE V. Finished measurements (in inches).

Size	1/2 Breast 1/	Sleeve length 2/	Back length 3/	Belt length 4/
Extra short				
32	20-1/8	17	42-1/8	42-1/2
34	20-7/8	17	42-3/8	44-1/2
36	21-7/8	17	42-5/8	46-1/2
38	22-7/8	17	42-7/8	48-1/2
40	23-7/8	17	43-1/8	50-1/2
42	24-7/8	17	43-3/8	52-1/2
Short				
30	19-1/8	18	43-3/8	40-1/2
32	20-1/8	18	43-5/8	42-1/2
34	20-7/8	18	43-7/8	44-1/2
36	21-7/8	18	44-1/8	46-1/2
38	22-7/8	18	44-3/8	48-1/2
40	23-7/8	18	44-5/8	50-1/2
42	24-7/8	18	44-7/8	52-1/2
44	25-7/8	18	45-1/8	54-1/2
46	26-7/8	18	45-3/8	56-1/2
Regular				
30	19-1/8	19	45-3/8	40-1/2
32	20-1/8	19	45-5/8	42-1/2
34	20-7/8	19	45-7/8	44-1/2
36	21-7/8	19	46-1/8	46-1/2
38	22-7/8	19	46-3/8	48-1/2
40	23-7/8	19	46-5/8	50-1/2
42	24-7/8	19	46-7/8	52-1/2
44	25-7/8	19	47-1/8	54-1/2
46	26-7/8	19	47-3/8	56-1/2
48	27-7/8	19	47-5/8	58-1/2

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TABLE V. Finished measurements (in inches) - Continued

Size	1/2 Breast 1/	Sleeve length 2/	Back length 3/	Belt length 4/
Long				
32	20-1/8	20	47-5/8	42-1/2
34	20-7/8	20	47-7/8	44-1/2
36	21-7/8	20	48-1/8	46-1/2
38	22-7/8	20	48-3/8	48-1/2
40	23-7/8	20	48-5/8	50-1/2
42	24-7/8	20	48-7/8	52-1/2
44	25-7/8	20	49-1/8	54-1/2
46	26-7/8	20	49-3/8	56-1/2
48	27-7/8	20	49-5/8	58-1/2
Extra long				
34	20-7/8	21	49-7/8	44-1/2
36	21-7/8	21	50-1/8	46-1/2
38	22-7/8	21	50-3/8	48-1/2
40	23-7/8	21	50-5/8	50-1/2
42	24-7/8	21	50-7/8	52-1/2
44	25-7/8	21	51-1/8	54-1/2
46	26-7/8	21	51-3/8	56-1/2
48	27-7/8	21	51-5/8	58-1/2
Tolerance				
Plus	1/2	1/2	1/2	1
Minus	1/2	1/2	1	1

1/ One half breast measurement shall be taken with coat buttoned from folded edge to folded edge in line with pit of armhole.

2/ Sleeve length measurement shall be taken along inseam from pit of armhole to bottom of sleeve.

3/ Back length measurement shall be taken along center seam from undercollar seam to bottom.

4/ Belt measurement shall be taken from point of finished belt to fold at buckle bar.

3.8 Workmanship. The finished raincoats shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.1.1 Component testing. Components shall be tested for the characteristics specified in FED-STD-191, wherever applicable, and as listed in table VII. All requirements are applicable to the sample unit. All test reports shall contain the individual values used in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified in table VII. The lot size and sample size shall be in accordance with the following:

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Lot size (lot unit)	Sample size (sample unit)
800 or less	2
801 to 22,000	3
22,001 and over	5

The basis for lot size and sample units for testing shall be in accordance with table VI.

TABLE VI. Basis for lot sizes and sample units for testing

Component	Lot expressed in	Sample unit for testing
Cloth, black, cotton plain weave or polyester spunlaced nonwoven (fusible)	Yards	1 yard full width
Fusible interlining fused to basic cloth	Yards	Two 24 inch by 24 inch specimens 1/
Tape, non-woven polyamide fusible web for seams	Yards	1/2 linear yard

1/ Prepared by the contractor during the same period and under the same conditions as for the fusing of the end item.

TABLE VII. Component testing

Component	Characteristic	Requirement paragraph	Test method
Cloth, black cotton, plain weave (fusible) or cloth, black polyester, spunlaced nonwoven (fusible)	Weight	3.2.3	5041
	Yarns per inch	3.2.3	5050
	Woven only		
	Breaking strength	3.2.3	5100
	Fiber content	3.2.3	1/
	Spunlaced construction	3.2.3	1/

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TABLE VII. Component testing - Continued

Component	Characteristic	Requirement paragraph	Test method
All dyed fusible interlinings	Colorfastness: - laundering	3.2.3.1	5610
Interlining fused to basic cloth	Bond strength: - initial	3.2.3.3	5951
	- after 3 launderings	3.2.3.3	4.3.1 and 5951
	- after 3 dry cleanings	3.2.3.3	5580 2/ and 5951
	Shrinkage: - after 1 laundering	3.2.3.4	4.3.1 3/
Polyamide fusible web for seams	Width	3.2.4	5020
	Weight	3.2.4	5041

1/ Certificate of compliance will serve as substitute for a test report.

2/ Pressing and shrinkage requirements shall not apply.

3/ Except that the specimens shall be subjected to only one cycle of laundering and drying. The specimens shall be prepared and the shrinkage calculated in accordance with Method 5556.

4.2.2 Point count inspection. Sampling and inspection provisions for end item examination, dimensional examination, and packaging inspection shall be performed in accordance with MIL-STD-1494.

4.2.3 End item testing. The finished coats shall be tested as specified in table VIII. The lot size shall be expressed in units of coats. The sample unit shall be one finished coat. The lot shall be unacceptable if one or more sample units fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (coats)	Sample size (sample units)
800 or less	2
801 to 22,000	3
22,001 and over	5

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TABLE VIII. End item testing

Characteristic	Requirement paragraph	Test method
Bond strength of fused areas after 3 launderings	3.2.3.5	4.3.1 and 5951 1/
Strike through of adhesive	3.2.3.5	Visual

1/ Method 5951 shall be followed except that the bond strength shall be based on the average of four specimens. It will be necessary to remove buttons and/or stitching to obtain the required number of specimens. When four fully laminated specimens cannot be obtained, partially or fully delaminated specimens shall be included in the average.

4.2.4 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.3 Methods of inspection.

4.3.1 Laundering procedure.

4.3.1.1 *Apparatus and material used for laundering.* The following apparatus and material are required for laundering:

a. Automatic Washing Machine (see 6.5).
b. Automatic Dryer (see 6.5).

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c. AATCC Standard Detergent 124 or equivalent.

d. Dummy pieces for ballast - 3 feet by 3 feet hemmed pieces of the basic material to be added to test coats or fused specimens to make a 4-pound load.

4.3.1.2 Procedure. If the coats or fused specimens weigh less than 4 pounds, ballast to make up a standard 4-pound load shall be added to the test coats or fused specimens. The ballast shall consist of pieces of the basic fabric. The pieces shall be hemmed and measure 3 feet by 3 feet. The washer shall be set for high water level, warm water temperature, and a 12-minute wash on the wash and wear cycle. The water used shall have a hardness not greater than 50 parts per million (ppm). The wash temperature shall be 140° ± 5°F. When the washer has filled, add 140 grams of detergent. Allow the washer to proceed automatically through the final spin cycle. Remove the coats or fused specimens immediately at the completion of the final spin and separate the coats or fused specimens and dummy pieces, when required, if tangled. Place the complete washed load in the dryer and dry at the normal cycle and at a setting which generates an exhaust temperature range of 140° to 160°F. Operate the dryer until the load is dry and continue tumbling 5 minutes with the heat turned off (cool-down cycle). Remove the load immediately after the machine stops. Repeat the wash and dry cycle until the load has completed three cycles.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each raincoat shall be buttoned, belted, and folded to measure approximately 15 inches by 11-1/2 inches. Each folded raincoat shall be inserted in a snug-fitting, flat style clear polyethylene film bag conforming to A-A-50083.

5.1.2 Commercial. Raincoats shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Twenty raincoats of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each fiberboard shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, alternated end for end, two in length, one in width, and ten in depth within a shipping container. Inside dimensions of

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each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.2.2. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Twenty raincoats of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW), or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, alternated end to end, two in length, one in width, and ten in depth within a shipping container. Inside dimensions of each shipping container shall be 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in PPP-B-636, except that the inspection shall be in accordance with 4.2.2.

5.2.3 Commercial packing. Raincoats, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), raincoats, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

5.4.2 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 inches by 4 inches with the words, "MIXED NSN's" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity of NSN's contained therein.

6. NOTES

6.1 Intended use. The raincoat is intended for wear by male military personnel of the Department of the Army.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Size required (see 1.2).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 Sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 Fusing conditions. The following fusing conditions were established to include recommended requirements of several fusible contractors. These limits are furnished for guidance only. For optimum results with the particular fusible used, specific machine setting should be obtained from the contractor furnishing the fusible material. Also conditions may vary depending on the fusing press type (tray, conveyor, or infrared).

	Cotton plain weave fusible		Polyester spunlaced non-woven fusible	
	(min)	(max)	(min)	(max)
Temperature (between pressing surfaces)	275°F	320°F	347°F	360°F
Pressure (between pressing surfaces), psi	2	4	75	85
Pressing time, seconds	8.0	12.0	8.0	9.0

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6.5 Recommended apparatus. The recommended apparatus for 4.3.1.1a and b are:

(1) Kenmore Automatic Washer Model 600 or similar machine. In case of dispute the Kenmore Model 600 shall be used.

(2) Kenmore Automatic Dryer Model 600 or equivalent.

6.6 Subject term (key word) listing.

Clothing, men's
Quarpel treated
Raincoat, men's

6.7 Supersession data. Class 1 raincoat and its requirements have been deleted since it is no longer required in the system.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

Project No. 8405-A043

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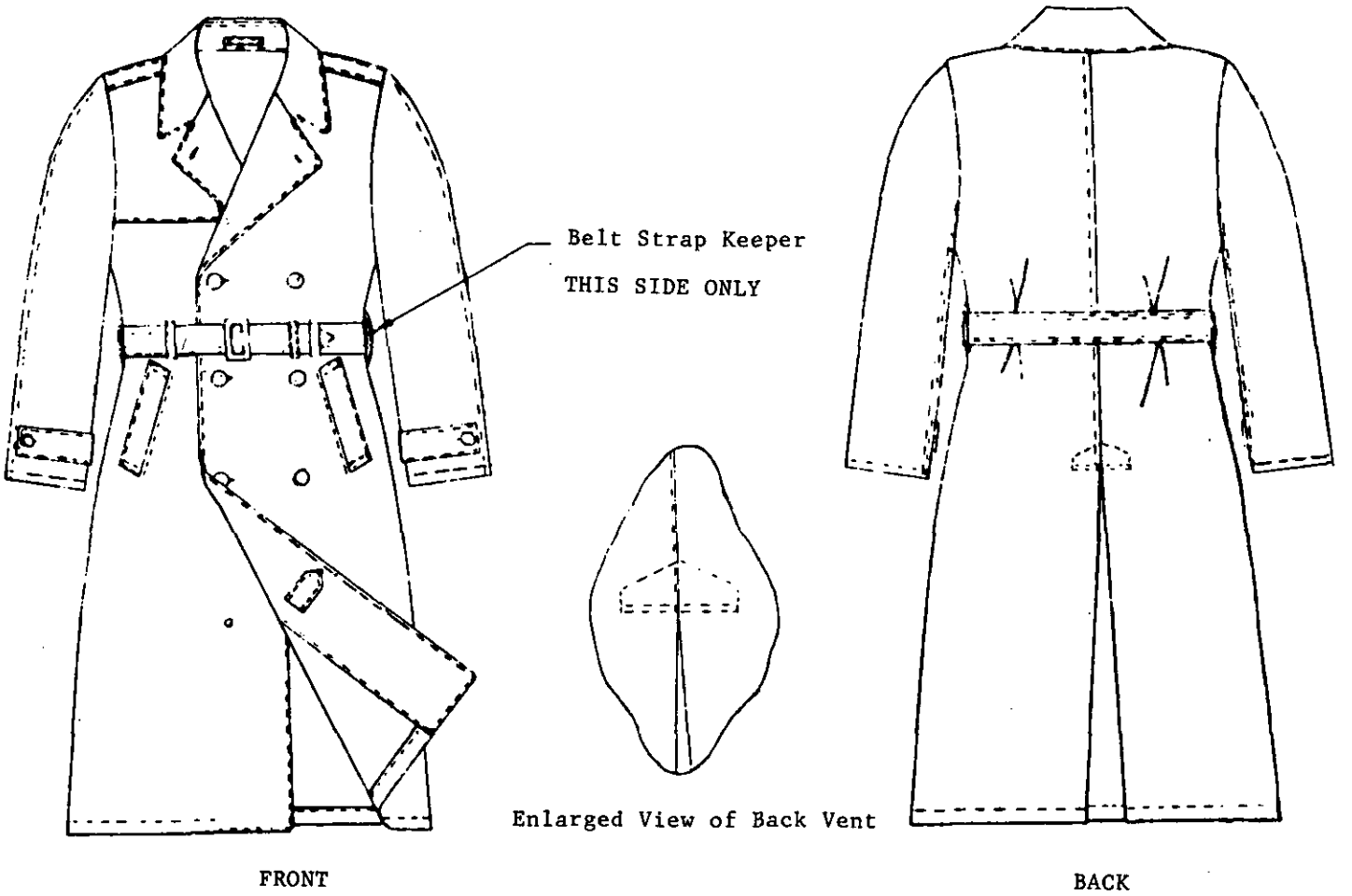


FIGURE 1. Raincoats, men's, quarpel.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-43518 C (GL)	Raincoats, Men's, Quarpel

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

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